Chapter 7

Reactions in Aqueous Solutions

Predicting Whether a Reaction Will Occur

- Why does a chemical reaction occur?
- As chemists have studied reactions, they have recognized several “tendencies” in reactants that drive them to form products.
  - That is, there are several “driving forces” that pull reactants toward products—changes that tend to make reactions go in the direction of the arrow.



Driving Forces

- The most common of these driving forces are

1. Formation of a solid
2. Formation of water
3. Transfer of electrons
4. Formation of a gas

Research Activity – Unit 7.1

- Research chemical reactions and find a chemical reaction that corresponds to each of the following categories:

1. Formation of a solid
2. Formation of water
3. Transfer of electrons
4. Formation of a gas

Precipitation Reactions

- One driving force for a chemical reaction is the formation of a solid, a process called precipitation.
- The solid that forms is called a precipitate, and the reaction is known as a precipitation reaction.



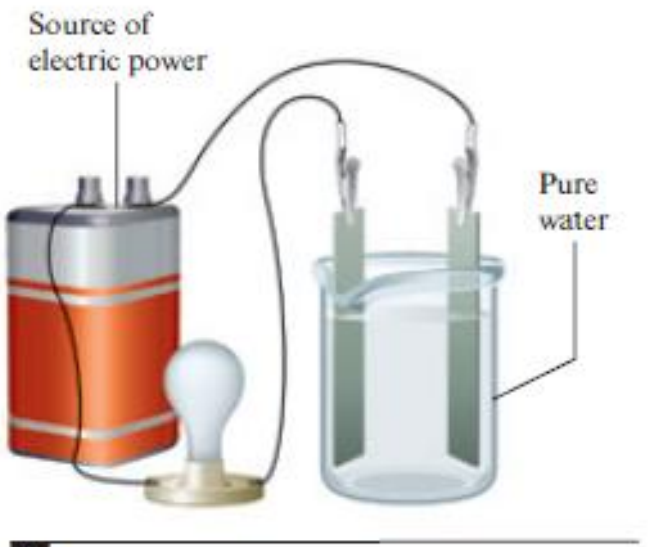
Ionic Compounds in Water

- In virtually every case when a solid containing ions dissolves in water, the ions separate and move around independently.
  - That is, $Ba(NO_3)_2(aq)$ does not contain $Ba(NO_3)_2$ units.
  - Rather, it contains separated Ba^{2+} and NO_3^- ions.
  - In the solution there are two NO_3^- ions for every Ba^{2+} ion.

Strong Electrolytes

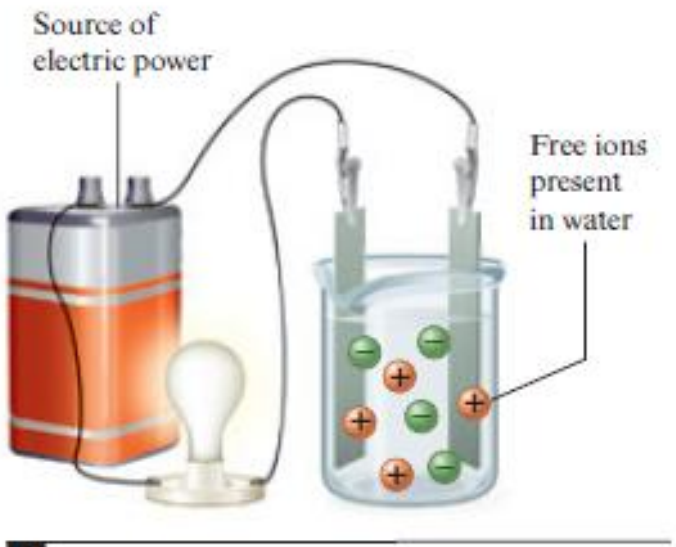
- Chemists know that separated ions are present when a solution is an excellent conductor of electricity.
- Pure water does not conduct an electric current.
- Ions must be present in water for a current to flow.
- When each unit of a substance that dissolves in water produces separated ions, the substance is called a strong electrolyte.

Electrolytes



A

Pure water does not conduct an electric current. The lamp does not light.



B

When an ionic compound is dissolved in water, current flows and the lamp lights. The result of this experiment is strong evidence that ionic compounds dissolved in water exist in the form of separated ions.

Mixing Electrolytes



$$2K^+(aq) + CrO_4^{2-}(aq) + Ba^{2+}(aq) + 2NO_3^-(aq) \rightarrow \text{Products}$$

The ions in $K_2CrO_4(aq)$

The ions in $Ba(NO_3)_2(aq)$

How to Decide What Products Form

- A solid compound must have a zero net charge.
  - This means that the product of our reaction must contain both anions and cations (negative and positive ions).
  - For example, K^+ and Ba^{2+} could not combine to form the solid because such a solid would have a positive charge. Similarly, CrO_4^{2-} and NO_3^- could not combine to form a solid because that solid would have a negative charge.
- Most ionic materials contain only two types of ions—one type of cation and one type of anion.

Solubility Rules

Table 7.1 General Rules for Solubility of Ionic Compounds (Salts) in Water at 25 °C

1. Most nitrate (NO_3^-) salts are soluble.
2. Most salts of Na^+, K^+, and NH_4^+ are soluble.
3. Most chloride salts are soluble. Notable exceptions are AgCl, $PbCl_2$, and Hg_2Cl_2.
4. Most sulfate salts are soluble. Notable exceptions are $BaSO_4$, $PbSO_4$, and $CaSO_4$.
5. Most hydroxide compounds are only slightly soluble.* The important exceptions are NaOH and KOH. $Ba(OH)_2$ and $Ca(OH)_2$ are only moderately soluble.
6. Most sulfide (S^{2-}), carbonate (CO_3^{2-}), and phosphate (PO_4^{3-}) salts are only slightly soluble.*

*The terms *insoluble* and *slightly soluble* really mean the same thing: such a tiny amount dissolves that it is not possible to detect it with the naked eye.

Solubility Practice

- Classify the following compounds as either soluble or insoluble:

$NaOH$ soluble

$BaSO_4$ insoluble

$CaCO_3$ insoluble

TopHat

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How to Predict Precipitates When Solutions of Two Ionic Compounds Are Mixed

1. Write the reactants as they actually exist before any reaction occurs.
   - Remember that when a salt dissolves, its ions separate.
2. Consider the various solids that could form.
   - To do this, simply exchange the anions of the added salts.
3. Use the solubility rules (Table 7.1) to decide whether a solid forms and, if so, to predict the identity of the solid.

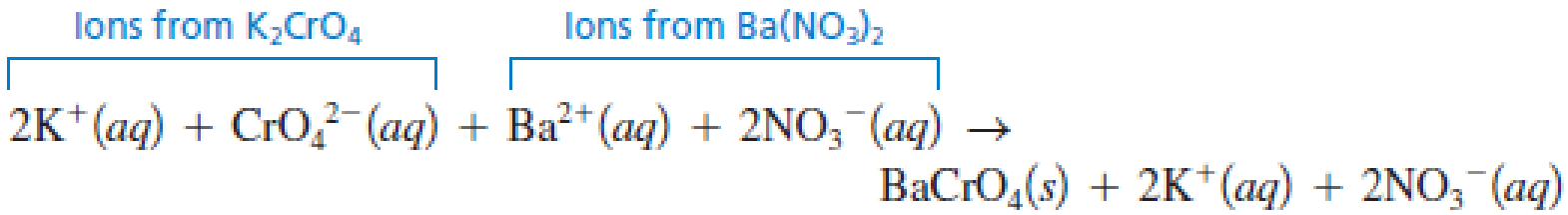
Molecular Equations

- A molecular equation shows the complete formulas of all reactants and products.
  - However, although this equation shows the reactants and products of the reaction, it does not give a very clear picture of what actually occurs in solution.

$$K_2CrO_4(aq) + Ba(NO_3)_2(aq) \rightarrow BaCrO_4(s) + 2KNO_3(aq)$$

Complete Ionic Equations

- A complete ionic equation better represents the actual forms of the reactants and products in solution.
  - In a complete ionic equation, all substances that are strong electrolytes are represented as ions.



Net Ionic Equations

- Ions which do not participate directly in a reaction in solution are called spectator ions.
- Net ionic equations include only those components that are directly involved in the reaction.
  - Chemists usually write the net ionic equation for a reaction in solution, because it gives the actual forms of the reactants and products and includes only the species that undergo a change.

$$Ba^{2+}(aq) + CrO_4^{2-}(aq) \rightarrow BaCrO_4(s)$$

Types of Equations for Reactions in Aqueous Solutions

- Three types of equations are used to describe reactions in solutions.

1. The molecular equation shows the overall reaction but not necessarily the actual forms of the reactants and products in solution.
2. The complete ionic equation represents all reactants and products that are strong electrolytes as ions. All reactants and products are included.
3. The net ionic equation includes only those components that undergo a change. Spectator ions are not included.

Writing Equations Practice

- Write the complete ionic and net ionic equation for the following reaction:

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$

Practice

- Write the molecular, complete ionic, and net ionic equation for the reaction of a solution of sodium chloride being added to a solution of silver nitrate.

Arrhenius Acids

- Svante Arrhenius proposed that an acid is a substance that produces H^+ ions (protons) when it is dissolved in water.
- Studies show that when HCl, HNO_3, and H_2SO_4 are placed in water, virtually every molecule dissociates to give ions.
- Because these substances are strong electrolytes that produce H^+ ions, they are called strong acids.

Arrhenius Bases

- Arrhenius also found that aqueous solutions that exhibit basic behavior always contain hydroxide ions.
- He defined a base as a substance that produces hydroxide ions (OH^-) in water.
- Hydroxide compounds that are strong electrolytes are called strong bases.

Acid-Base Reactions

- There are two important things to note as we examine the reaction of hydrochloric acid with aqueous sodium hydroxide and the reaction of nitric acid with aqueous potassium hydroxide.

1. The net ionic equation is the same in both cases; water is formed.

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

2. Besides water, which is always a product of the reaction of an acid with OH^-, the second product is an ionic compound (called a salt), which might precipitate or remain dissolved, depending on its solubility.

$$HCl(aq) + NaOH(aq) \rightarrow H_2O(l) + NaCl(aq)$$
$$HNO_3(aq) + KOH(aq) \rightarrow H_2O(l) + KNO_3(aq)$$

Dissolved ionic compounds

Summary of Strong Acids and Strong Bases – Principles 1-2

- The following points about strong acids and strong bases are particularly important.

1. The common strong acids are aqueous solutions of HCl, HNO_3, and H_2SO_4.
2. A strong acid is a substance that completely dissociates (ionizes) in water. (Each molecule breaks up into an H^+ ion plus an anion.)

Summary of Strong Acids and Strong Bases – Principles 3-4

- The following points about strong acids and strong bases are particularly important.

3. A strong base is a metal hydroxide compound that is very soluble in water. The most common strong bases are NaOH and KOH, which completely break up into separated ions (Na^+ and OH^- or K^+ and OH^-) when they are dissolved in water.
4. The net ionic equation for the reaction of a strong acid and a strong base (contains OH^-) is always the same: it shows the production of water.

Summary of Strong Acids and Strong Bases – Principles 5-6

- The following points about strong acids and strong bases are particularly important.

5. In the reaction of a strong acid and a strong base, one product is always water, and the other is always an ionic compound called a salt, which remains dissolved in the water. This salt can be obtained as a solid by evaporating the water.
6. The reaction of H^+ and OH^- is often called an acid–base reaction, where H^+ is the acidic ion and OH^- is the basic ion.

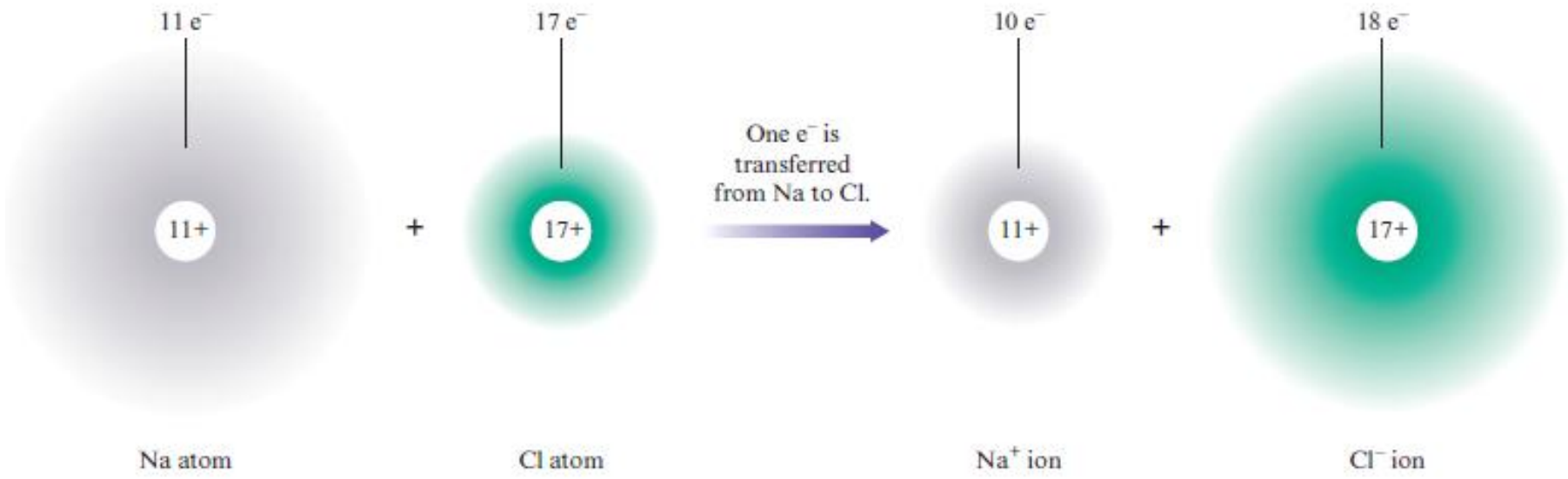
Practice

- Balance the reaction, write the complete and net ionic equations.
  - $H_3PO_4 + Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + H_2O$

- Write the molecular reaction of nitrous acid and potassium hydroxide and balance it. Show the complete and net ionic equations.

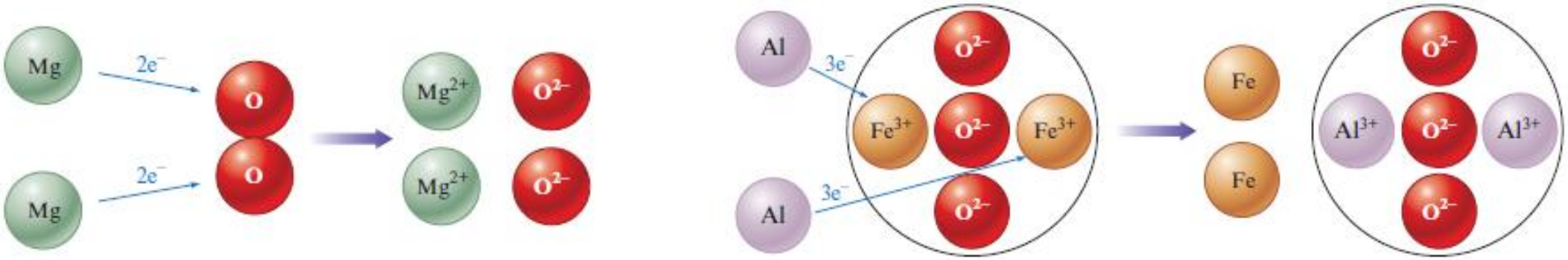
Transferring Electrons

- When a sodium atom reacts with a chlorine atom an electron is transferred.



Electron Transfer in Reactions

- A reaction that involves a transfer of electrons is called an oxidation–reduction reaction.



Characteristics of Oxidation–Reduction Reactions – Principle 1

1. When a metal reacts with a nonmetal, an ionic compound is formed.
   - The ions are formed when the metal transfers one or more electrons to the nonmetal, the metal atom becoming a cation and the nonmetal atom becoming an anion.
   - Therefore, a metal–nonmetal reaction can always be assumed to be an oxidation–reduction reaction, which involves electron transfer.

Characteristics of Oxidation–Reduction Reactions – Principle 2

2. Two nonmetals can also undergo an oxidation–reduction reaction.
   - At this point we can recognize these cases only by looking for O_2 as a reactant or product.
   - When two nonmetals react, the compound formed is not ionic.

Oxidation–Reduction Reactions Practice

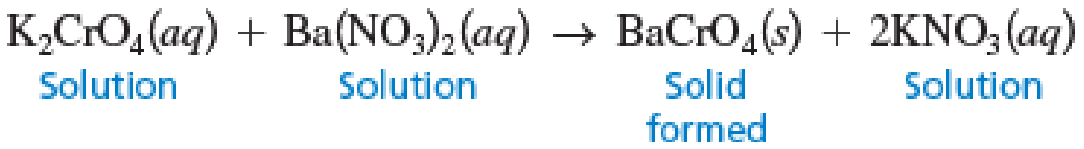
- Is the following an oxidation–reduction reaction? If so, how many electrons are transferred from each Al atom?

  $2Al + N_2 \rightarrow 2AlN$

Practice

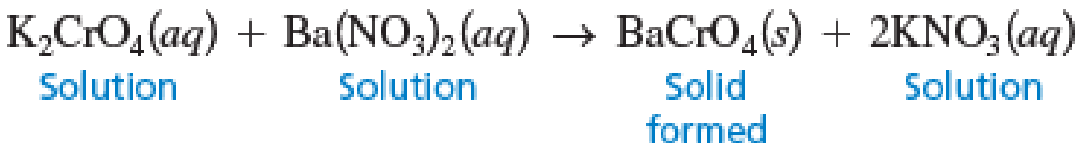
- Solid copper reacts with aqueous iron(II) chloride to form solid iron and aqueous copper(II) chloride.
- Write the molecular equation, balance it, and determine how many electrons are lost or gained by the various species.

Precipitation Reaction



- In this reaction solid $BaCrO_4$ (a precipitate) is formed.
- Because the formation of a solid when two solutions are mixed is called precipitation, we call this a precipitation reaction.

Double-Displacement Reactions



- Notice in this reaction that two anions (NO_3^- and CrO_4^{2-}) are simply exchanged.
- Note that CrO_4^{2-} was originally associated with K^+ in K_2CrO_4 and that NO_3^- was associated with Ba^{2+} in $Ba(NO_3)_2$.
- In the products these associations are reversed.
- Because of this double exchange, we sometimes call this reaction a double-exchange reaction or double-displacement reaction.

Acid–Base Reactions

$$HCl(aq) + KOH(aq) \rightarrow H_2O(l) + KCl(aq)$$

- We classify this reaction as an acid–base reaction.
- You can recognize this as an acid–base reaction because it involves an H^+ ion that ends up in the product water.

Oxidation–Reduction Reactions

$$2Li(s) + F_2(g) \rightarrow 2LiF(s)$$

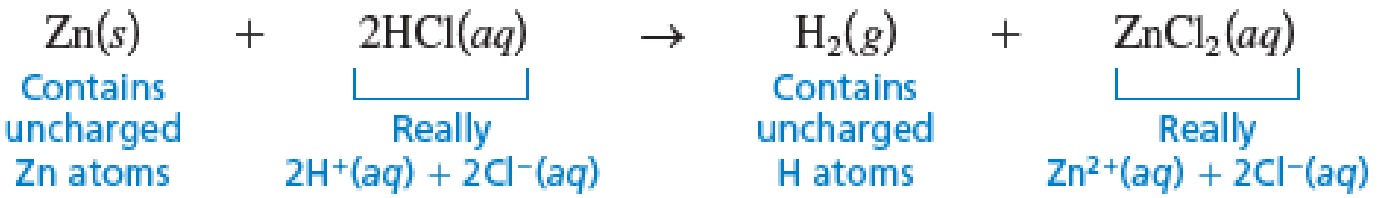
- In this reaction each lithium atom loses one electron to form Li^+, and each fluorine atom gains one electron to form the F^- ion.
- The process of electron transfer is also called oxidation–reduction.
- Thus, we classify the preceding reaction as an oxidation–reduction reaction.

Gas Evolution Reactions

$$2HCl(aq) + Na_2CO_3(aq) \rightarrow CO_2(g) + H_2O(l) + NaCl(aq)$$

- In this reaction carbon dioxide gas is formed.
- This reaction is also an acid–base reaction.

Single-Replacement Reactions



- This reaction is an oxidation–reduction reaction.
- This reaction is also a single-replacement reaction since a single type of anion (Cl^-) was exchanged between H^+ and Zn^{2+}.

Combustion Reactions

- Many chemical reactions that involve oxygen produce energy (heat) so rapidly that a flame results.
- Such reactions are called combustion reactions.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

Synthesis (Combination) Reactions

- When a given compound is formed from simpler materials, we call this a synthesis (or combination) reaction.

$$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$$

$$Mg(s) + F_2(g) \rightarrow MgF_2(s)$$

Decomposition Reactions

- In many cases a compound can be broken down into simpler compounds or all the way to the component elements.
  - This is usually accomplished by heating or by the application of an electric current.
- Such reactions are called decomposition reactions.

$$2H_2o(l) \xrightarrow[\text{Electric current}]{} 2H_2(g) + O_2(g)$$

$$2HgO(s) \xrightarrow[\text{Heat}]{} 2Hg(l) + O_2(g)$$

Reaction Classification Practice

- Classify the following as combustion, synthesis, or decomposition reactions:

$N_2 + 3H_2 \rightarrow 2NH_3$

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

$2H_2O \rightarrow 2H_2 + O_2$

Summary of Classes of Reactions

